Machine Learning-Based Prediction of California Bearing Ratio (CBR) of Soils

A Closed-Loop Geotechnical Framework: Conformal Uncertainty & Minimum-Test Verification Engine

1. Executive Overview & Research Problem

In pavement and highway engineering, the **California Bearing Ratio (CBR)** governs subgrade thickness design. Standard laboratory testing per **ASTM D1883-16** mandates a **96-hour (4-day) water-soaking period** followed by mechanical penetration. This empirical process is labor-intensive, time-consuming, and expensive. While traditional machine learning models predict CBR from index properties, they suffer from two critical deficiencies in civil practice: (1) *black-box deterministic point estimates without safety guarantees*, and (2) *an inability to guide the engineer on what to do when predictions are uncertain*.

Project Contribution: We develop a safety-critical geotechnical AI system that not only delivers high-accuracy CBR predictions, but also computes **90% Conformal Prediction Confidence Intervals**, evaluates a composite **Soil Reliability Index (SRI)**, and introduces a novel **Minimum-Test Verification Engine** that advises the laboratory which test to execute to minimize uncertainty at minimal cost and time.

2. Experimental Dataset Characteristics (ASTM D1883-16)

Database Source: Nagaraju et al., *Predicting California Bearing Ratio of Lateritic Soils Using Hybrid Machine Learning Technique*, **Buildings 2023**, 13(1), 255. Extracted from Table A1 (149 experimental lateritic soil specimens).

Measured Soil Parameter	Standard / Source	Min	Mean	Max	Geotechnical Role in Subgrade
Gravel Content (%)	ASTM D422 (Sieve)	0.0%	22.65%	65.0%	Coarse aggregate skeleton & mechanical interlocking
Sand Content (%)	ASTM D422 (Sieve)	14.0%	42.28%	88.0%	Frictional angle component and granular packing
Fines Content (Silt+Clay, %)	Passing 75 µm sieve	8.0%	35.20%	78.0%	Cohesion, moisture sensitivity, swell potential
Liquid Limit (LL, %)	ASTM D4318	0.0%	33.56%	68.0%	Compressibility and water absorption boundary
Plastic Limit (PL, %)	ASTM D4318	0.0%	19.33%	27.0%	Lower moisture limit of plastic consistency
Plasticity Index (PI, %)	PI = LL - PL	0.0%	14.22%	41.0%	Degree of clay activity and sensitivity to water
Optimum Moisture Content (OMC, %)	ASTM D698 (Proctor)	5.30%	13.25%	30.70%	Compaction moisture control benchmark
Max Dry Density (MDD, g/cc)	ASTM D698 (Proctor)	1.870	2.080	2.280	Dry unit weight; controls void ratio & soil density
Target Soaked CBR (%)	**ASTM D1883-16**	**12.0%**	**25.42%**	**34.0%**	**Primary highway pavement structural design metric**

USCS Classification Representation: Clayey Sand (SC: 31.5%), Clayey Gravel (GC: 29.5%), Lean Clay (CL: 22.1%), Silty Sand (SM: 12.1%), Fat Clay (CH: 3.4%), and Gravel (GW/GP: 1.3%). This balanced distribution represents both granular/frictional and cohesive subgrades commonly encountered in infrastructure corridors.

3. Multi-Model Experimental Benchmarking (5-Fold Cross-Validation)

Five model architectures were evaluated using strict 5-fold cross-validation on the experimental dataset. Hyperparameters were cross-validated to avoid overfitting:

Machine Learning Algorithm	R^2 (Mean ± Std)	RMSE (CBR %)	MAE (CBR %)	Geotechnical Characteristics & Behavior
Multiple Linear Regression (MLR)	**0.9736 ± 0.011**	**0.781 ± 0.221**	**0.592 ± 0.098**	High parametric fidelity; strong linear dependence on dry density & gravel.
Artificial Neural Network (ANN)	0.9639 ± 0.009	0.919 ± 0.196	0.687 ± 0.107	Multi-Layer Perceptron (64-32 neurons) with ReLU; non-linear mapping.
XGBoost Regressor (XGB)	0.9466 ± 0.014	1.104 ± 0.190	0.833 ± 0.089	Gradient boosted trees; robust to collinearity between gradation variables.
Support Vector Regression (SVR)	0.9434 ± 0.029	1.137 ± 0.362	0.784 ± 0.146	Radial Basis Function (RBF) kernel; high boundary stability.
Random Forest Regressor (RF)	0.9338 ± 0.021	1.228 ± 0.266	0.873 ± 0.098	Ensemble bagging (100 trees); non-parametric, low variance primary backbone.

4. Phase IX: Conformal Uncertainty & Soil Reliability Index (SRI)

Standard ML outputs point values that do not convey risk. We implemented a rigorous safety engine:

- **Conformal Prediction (90% Guarantee):** Computes distribution-free residual quantiles on held-out calibration data, establishing a calibrated margin of **±1.46 CBR units** (Mean Prediction Interval Width = 3.21 units).
- **Soil Reliability Index (SRI, 0–100%):** Fuses 3 diagnostic metrics into a unified index:
  SRI = 0.35 × (Model Disagreement) + 0.40 × (Interval Sharpness) + 0.25 × (Domain Proximity)
- **Safety Verification:** Samples labeled HIGH_CONFIDENCE (SRI ≥ 75%) exhibited an average test error of only **0.56 CBR units**, whereas uncertain samples exhibited **2.22 CBR units**, proving the engine flags risky predictions reliably.

5. Phase X: Minimum-Test Verification Engine & Test Selection Score (TSS)

Core Research Novelty: When a soil sample has high uncertainty (SRI < 75%), running a full 4-day soaked CBR test is slow and costly. Our engine computes the **Test Selection Score (TSS)** to identify the laboratory test providing maximum uncertainty reduction per unit burden:

$$\mathbf{TSS_i = \Delta U_i \,/\, (\alpha \cdot Cost_i + \beta \cdot Time_i + \gamma \cdot Effort_i)} \quad \textit{[Operational weights: } \alpha\textit{=0.40, } \beta\textit{=0.35, } \gamma\textit{=0.25]}$$

Candidate Laboratory Test	Features Measured	Lab Time	Uncertainty Drop (ΔU)	Burden Score	TSS Rank & Recommendation
Grain Size / Sieve Analysis	Gravel, Sand, Fines (%)	4.0 Hours	**3.780 CBR units**	0.383	**9.857 (Rank #1 Recommended)**
Standard Proctor Compaction	OMC (%), MDD (g/cc)	24.0 Hours	0.289 CBR units	1.000	0.289 (Rank #2)
Atterberg Limits	LL, PL, PI (%)	2.0 Hours	0.024 CBR units	0.225	0.105 (Rank #3)

Geotechnical Finding: In lateritic soils, aggregate interlock governs bearing resistance. Sieve analysis yields nearly **13x greater uncertainty reduction** than compaction tests, making it the highest ROI test.

6. Phase XI & XII: Interactive Closed-Loop Recalibration Workflow

We simulated the active laboratory feedback loop on held-out specimens. Below is the exact progression for Specimen #122 (Clayey Gravel, True Soaked CBR = 33.0%):

Iteration	Known Test Inputs	Predicted CBR	90% Interval	SRI Score	Engine Decision & Engineering Action
Iteration 1	Atterberg Limits only (LL, PL, PI)	26.58%	[23.7%, 29.5%]	35.8% (Low)	REJECT PREDICTION; Engine triggers Sieve Analysis (TSS = 9.86)
Iteration 2	+ Sieve Analysis added (G, S, F)	31.69%	[29.4%, 34.0%]	71.4% (Moderate)	Interval narrows; absolute error drops from 6.42 to 1.31 units.
Iteration 3	+ Proctor Compaction (OMC, MDD)	**32.09%**	**[30.2%, 34.0%]**	**75.9% (High)**	**ACCEPT PREDICTION; Final Error = 0.91 units! (Target 33.0%)**

7. Academic Uniqueness (Comparison with Published Literature)

Evaluation Dimension	Conventional Machine Learning Papers	Our Engineered Closed-Loop Decision Framework
Prediction Format	Deterministic point value (e.g., 'CBR = 24.3%').	**Conformal 90% bounds [CBR_min, CBR_max] + Point estimate.**
Risk & Safety Control	None; unquantified risk of highway subgrade rutting/failure.	**Soil Reliability Index (SRI) alerts engineers when to accept predictions.**
Handling Incomplete Data	Requires all 8 tests upfront or fails completely.	**Gracefully models partial inputs; quantifies exact penalty of missing tests.**
Laboratory Guidance	None; purely descriptive academic exercise.	**Prescriptive TSS Engine optimizes testing schedule, cost, and technician time.**
Software Implementation	Static offline scripts or static tables.	**Interactive Streamlit web dashboard with real-time sliders and simulator.**

8. Key Summary Points for Oral Defense & Professor Q&A;

1. **Rigorous Experimental Grounding:** Validated on 149 experimental lateritic soil specimens from published civil engineering literature (Nagaraju et al., *Buildings 2023*) tested strictly under ASTM D1883-16.
2. **High Predictive Fidelity:** 5-fold cross-validation achieved R^2 up to 0.97 with mean absolute error of only 0.59 to 0.87 CBR units across five competitive machine learning families.
3. **Conformal Safety Guarantees:** Unlike standard black-box AI, conformal prediction intervals provide a distribution-free 90% confidence envelope, eliminating guesswork for highway pavement designers.
4. **Actionable Laboratory Economy:** The Minimum-Test Engine replaces arbitrary test planning with an optimal mathematical score (TSS = 9.86 for Sieve), saving up to 4 days of soaking and compaction time.